

Index

Unit – 4B $\rightarrow$ Land, Noise & Thermal Pollution	
1) Land Pollution.....	2
2) Land Use	2
3) Land Pollution & Causes	3
4) Land Degradation & Causes	3
5) Effects of Land Pollution.....	4
6) Control of Land Pollution	5
7) Noise Pollution	6
8) Sound & Noise	6
9) Sources of Noise Pollution	7
10) Effects of Noise Pollution	7
11) Control of Noise Pollution	7
12) Thermal Pollution	9
13) Sources of Thermal Pollution	9
14) Effects of Thermal Pollution	9

Prepared by: Prof. Arjun Chavada, DU

Unit – 4B $\rightsquigarrow$ Land, Noise & Thermal Pollution

Land Pollution

- Land is considered most important resource of the earth as it is source of many materials essential to man and other organisms and is essential medium for development of agriculture, forestry, vegetation, etc.
- It is major constitute of lithosphere; it forms about one fifth of the earth surface.
- The outer crust of the earth is called lithosphere.
- Living organism, plant and vegetation are supported by lithosphere.

Land Use

- Most human or natural activities need space for their location and development, this space is provided by land.
- Various land uses are as following;
 1. Food production
 2. Shelter, living space
 3. Water resources & other raw material (minerals and ores)
 4. Industrial purposes
 5. Commercial purposes
 6. Waste disposal
 7. Energy purposes
- Though the pattern of land use varies from country to country; broadly the pattern of land use on earth is;
 - Agricultural land – 11%
 - Pastures and meadows – 22%
 - Forest land – 30 %
 - Urban and non-agricultural area – 37% (i.e., land area occupied by human dwelling and factories roads and railways, deserts, glaciers, polar ice, marshes, rocks and mountains)

Unit 3 – Environmental Pollution

- In India more than two-fifth of the land is agricultural land. The pattern of land use is as following;
- Agricultural land – 43.6%
 - Permanent pastures and meadows – 14.6%
 - Culturable wasteland – 12.2%
 - Forest – 10.7%
 - Barren and unculturable land- 8.4%
 - Urban land – 5.3%
 - No information available – 5.2%

Land pollution and its causes

- Land pollution is the degradation or contamination of the land surface of the earth.
- Causes of land pollution are;
1. Careless and uncontrolled dumping of urban waste
 2. Dumping of industrial waste
 3. Uncontrolled mining
 4. Misusing the soil by harmful agricultural practices
- Land pollution can be due to solid waste (industrial waste, mining waste, ashes, agricultural waste) and due to soil pollution or degradation.

Land degradation and its causes (over land use)

- The surface layer of the land, which is termed soil, is responsible for its productive capacity.
- This soil contains minerals which is utilized by plant for growth.
- The factors which deplete the mineral content of the top layer or damage it, reduce the fertility of the soil, is termed land pollutants.
- Land degradation refers to this deterioration or loss of fertility or productive capacity of the soil.
- Factors causing land degradation are;
1. Soil erosion

Unit 3 – Environmental Pollution

2. Soil pollution
3. Salination and water logging
4. Shifting cultivation
5. Desertification
6. Urbanization

Soil erosion:

- Loss or removal of the superficial layer of the soil by the action of water, wind or by the activity of man is termed as soil erosion.
- Factors affecting soil erosion are;
 1. Distribution, intensity and amount of rainfall: higher rainfall will result in more erosion of the soil.
 2. Slope of the ground: steeper grounds are prone to erosion compared to flat terrains
 3. Nature of soil: erosion will be more in open soil than heavier loam.
 4. Vegetation cover: vegetative cover reduces the erosion
 5. Soil management: faulty methods of drainage, over grazing, wrong method of cultivation affects erosion potential.

Effects of Land pollution

- Land pollution can lead to following adverse effects;
 1. Adverse effects on properties of soil.
 2. Reduced fertility and proportional reduction in productivity and quality of the agricultural product
 3. Salination: salination implies increase in the number of salts in soil.
 4. Salination is generally observed in arid region
 5. Desertification: Conversion of fertile land into non-fertile or desert
 6. It can be due to winds blown soil deposit
 7. Excessive grazing
 8. Migration for better life style
 9. Waste of available land use

Control of Land pollution

→ Control measures include;

1. Industries should be forced to develop their own effluent treatment plants (ETP) to minimize soil contamination due to toxic chemicals from industries.
2. Solid waste disposal should be managed properly and dumping should not be done in open spaces.
 - **Sanitary land filling** and methods like composting must be used for disposal of solid waste.
3. Excessive use of chemical fertilizers and pesticides should be avoided and **use of natural fertilizer** should be promoted.
4. Crop rotations should be practiced.
5. Afforestation can reduce the soil erosion and maintains nutrient circulation.

Noise pollution

Sound and Noise

- Sound is a physical phenomenon that stimulates the sense of hearing in human beings.
- Hearing takes place whenever vibration of the frequencies between about 15 to 20000 hertz reaches the inner ear.
- Sound is measured in decibels(dB) (sound pressure level).
- Sound pressure is the amount of air pressure fluctuation created by source of the sound.
- We hear or perceive sound pressure as loudness.
- An unwanted and unpleasant sound which may cause discomfort is called noise and presence of such sound in the atmosphere is called noise pollution.
- The World Health Organization (WHO) defines noise above 65 decibels (dB) as noise pollution.



Sound

Sound is defined as a type of energy that causes our ears to vibrate and produce the experience of hearing. It spreads in the shape of a wave.



Noise

Noises are defined as frequent indicators of unwanted sounds. A loud noise is heard when something bursts, people shout, something is banged.

Sources of Noise pollution

→ Constructional and Industrial machinery:

- It includes different types of equipment used in industries, generators, boring and drilling machineries, pile drivers, simple and pneumatic road rollers and other such machinery.

→ Means of Transport:

- This contains the sound created by means of different transport like aircraft, trains, tractors, trucks and other vehicles.

→ Other sources:

- it includes the sound generated from the use of loud speakers during marriages, festivals, political meetings, etc. Use of crackers during festivals and celebrations also creates high volume sound

Effects of Noise pollution

→ Major effects of noise pollution are

1. Loss of hearing
2. Annoyance
3. Health effects
4. Interference with communication
5. Working efficiency

→ Long time exposure of loud sound (80-90 dBA for more than eight hours a day) may cause loss in hearing and in chronic cases, the it may result in total deafness.

→ Sudden loudness particularly from crackers and explosions may affect the ear drum and sensitive ear membranes.

→ Physiological effects like headache, dizziness, high rate of heart beat.

→ Among different psychological effects some are depression and fatigue, insomnia, mental stress, behavioral changes.

Control of Noise pollution

→ People can use ear protection aids like ear plugs, noise helmets, headphones, etc., to reduce occupational exposure of noise.

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- Proper design and fabrication to replace noisy machines, lubrication and time to time servicing of machines, using sound absorbing materials and sound proof cabins, using silencers to control noise from automobiles and ducts, etc.
- Sound insulation:
 - Sound insulation can be done in building and offices by using acoustical tiles, perforated plywood, wooden ceiling and floors, sound absorbing materials in doors and windows.
- Planting green trees along the roads and in and around buildings also reduces noise.
- Regulations:
 - Strict rules and regulations like minimum use of loud speakers, restricting the use of loud horns, etc.

Thermal Pollution

Introduction:

- Thermal pollution refers to degradation of water quality as a result of any process that changes the ambient water temperature.
- A common cause of thermal pollution is use of water as a coolant by power plants (thermal and nuclear) and industries.
- Water used as coolant is returned to water sources like river, lakes, etc., at a higher temperature.

Sources of Thermal pollution

- Various thermal pollution are as follows:

1. Nuclear power plant
2. Coal fired power plant
3. Industrial effluent
4. Hydroelectric power plant
5. Domestic wastewater

Control of Thermal pollution

- Cooling towers
- Cooling ponds:
 - using cooling ponds to let the heated water lose its temperature with time.
- Evaporation pond:
 - spreading heated liquid in large open area to evaporate it with the help of sunlight.